

```
In [1]: import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

from sklearn.ensemble import RandomForestRegressor
```

```
In [2]: import os

print(os.getcwd())      # current folder
print(os.listdir())     # files in current folder
```

Out:

```
C:\Users\USER\OneDrive\Desktop\movie recom
['.ipynb_checkpoints', 'IMDb Movies India.csv.zip', 'Untitled.ipynb']
```

```
In [3]: import pandas as pd

df = pd.read_csv("IMDb Movies India.csv.zip", encoding="latin1")

df.head()
```

Out:

	Name	Year	Duration	Genre	Rating	Votes	Director	Actor 1	Actor 2	Actor 3
0		NaN	NaN	Drama	NaN	NaN	J.S. Randhawa	Manmauji	Birbal	Rajendra Bhatia
1	#Gadhvi (He thought he was Gandhi)	(2019)	109 min	Drama	7.0	8	Gaurav Bakshi	Rasika Dugal	Vivek Ghamande	Arvind Jangid
2	#Homecoming	(2021)	90 min	Drama, Musical	NaN	NaN	Soumyajit Majumdar	Sayani Gupta	Plabita Borthakur	Roy Angana
3	#Yaaram	(2019)	110 min	Comedy, Romance	4.4	35	Ovais Khan	Prateik	Ishita Raj	Siddhant Kapoor
4	...And Once Again	(2010)	105 min	Drama	NaN	NaN	Amol Palekar	Rajat Kapoor	Rituparna Sengupta	Antara Mali

```
In [4]: import os
        print(os.listdir())
```

Out:
['.ipynb_checkpoints', 'IMDb Movies India.csv.zip', 'Untitled.ipynb']

```
In [5]: import zipfile

        with zipfile.ZipFile('IMDb Movies India.csv.zip', 'r') as zip_ref:
            zip_ref.extractall()

        print("Extracted Successfully")
```

Out:
Extracted Successfully

```
In [6]: import os

        print(os.listdir())
```

Out:
['.ipynb_checkpoints', 'IMDb Movies India.csv', 'IMDb Movies India.csv.zip', 'Untitled.ipynb']

```
In [7]: import pandas as pd

        df = pd.read_csv("IMDb Movies India.csv", encoding="latin1")

        df.head()
```

Out:

	Name	Year	Duration	Genre	Rating	Votes	Director	Actor 1	Actor 2	Actor 3
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	Name	Year	Duration	Genre	Rating	Votes	Director	Actor 1	Actor 2	Actor 3
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4	...And Once Again	(2010)	105 min	Drama	NaN	NaN	Amol Palekar	Rajat Kapoor	Rituparna Sengupta	Antara Mali

```
In [8]: df.info()
```

```
Out:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 15509 entries, 0 to 15508
Data columns (total 10 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   Name        15509 non-null object
1   Year        14981 non-null object
2   Duration    7240 non-null  object
3   Genre       13632 non-null object
4   Rating      7919 non-null  float64
5   Votes       7920 non-null  object
6   Director    14984 non-null object
7   Actor 1     13892 non-null object
8   Actor 2     13125 non-null object
9   Actor 3     12365 non-null object
dtypes: float64(1), object(9)
memory usage: 1.2+ MB
```

```
In [9]: print(df.columns)
```

```
Out:
Index(['Name', 'Year', 'Duration', 'Genre', 'Rating', 'Votes', 'Director',
      'Actor 1', 'Actor 2', 'Actor 3'],
      dtype='object')
```

```
In [10]: print(df.columns.tolist())
```

```
Out:
['Name', 'Year', 'Duration', 'Genre', 'Rating', 'Votes', 'Director', 'Actor 1', 'Actor 2', 'Actor 3']
```

```
In [11]: df = df[['Name','Year','Duration','Genre',
                  'Rating','Votes','Director',
                  'Actor 1','Actor 2','Actor 3']]
```

```
In [12]: print(df.isnull().sum())
```

Out:

```
Name          0
Year           528
Duration      8269
Genre         1877
Rating        7590
Votes         7589
Director       525
Actor 1        1617
Actor 2        2384
Actor 3        3144
dtype: int64
```

```
In [13]: df['Year'] = df['Year'].astype(str)

df['Year'] = df['Year'].str.extract(r'(\d{4})')

df['Year'] = pd.to_numeric(df['Year'], errors='coerce')
```

```
In [14]: df['Duration'] = df['Duration'].astype(str)

df['Duration'] = df['Duration'].str.extract(r'(\d+)')

df['Duration'] = pd.to_numeric(df['Duration'], errors='coerce')
```

```
In [15]: df['Votes'] = df['Votes'].astype(str)

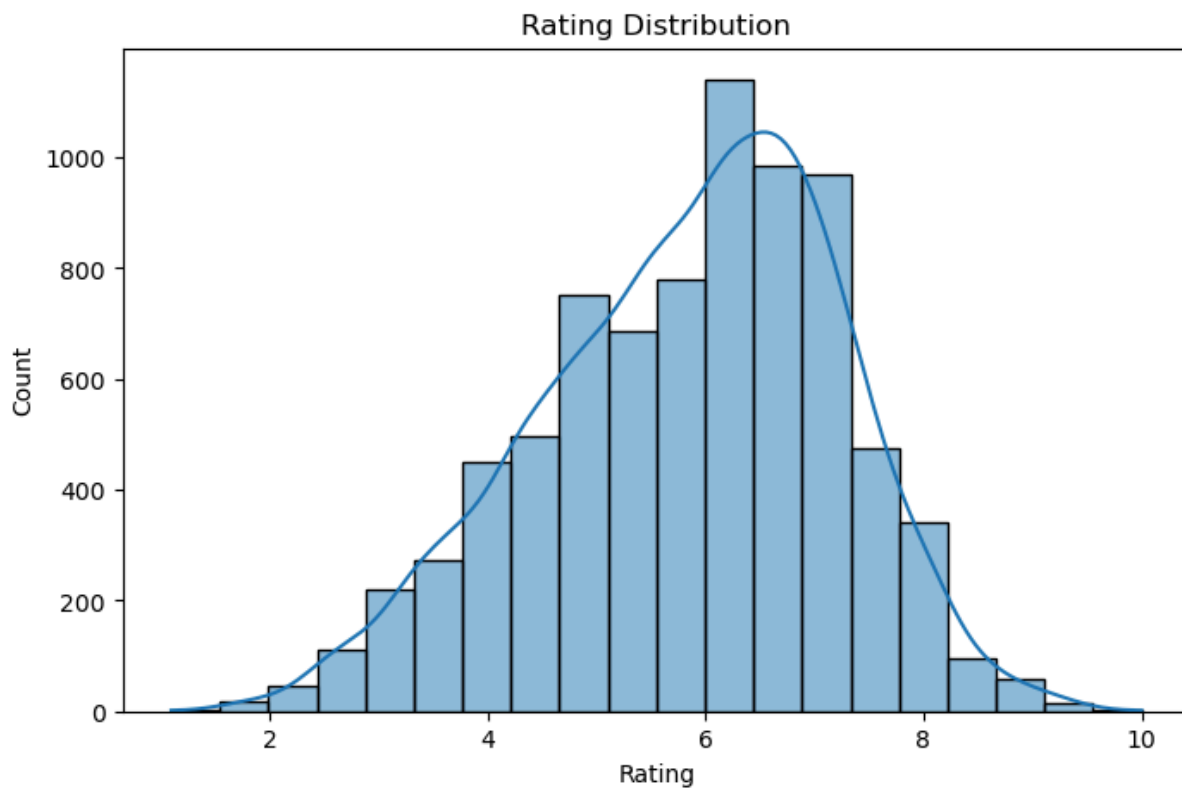
df['Votes'] = df['Votes'].str.replace(',', '')

df['Votes'] = pd.to_numeric(df['Votes'], errors='coerce')
```

```
In [16]: import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [17]: plt.figure(figsize=(8,5))
sns.histplot(df['Rating'], bins=20, kde=True)
plt.title("Rating Distribution")
plt.show()
```

Out:

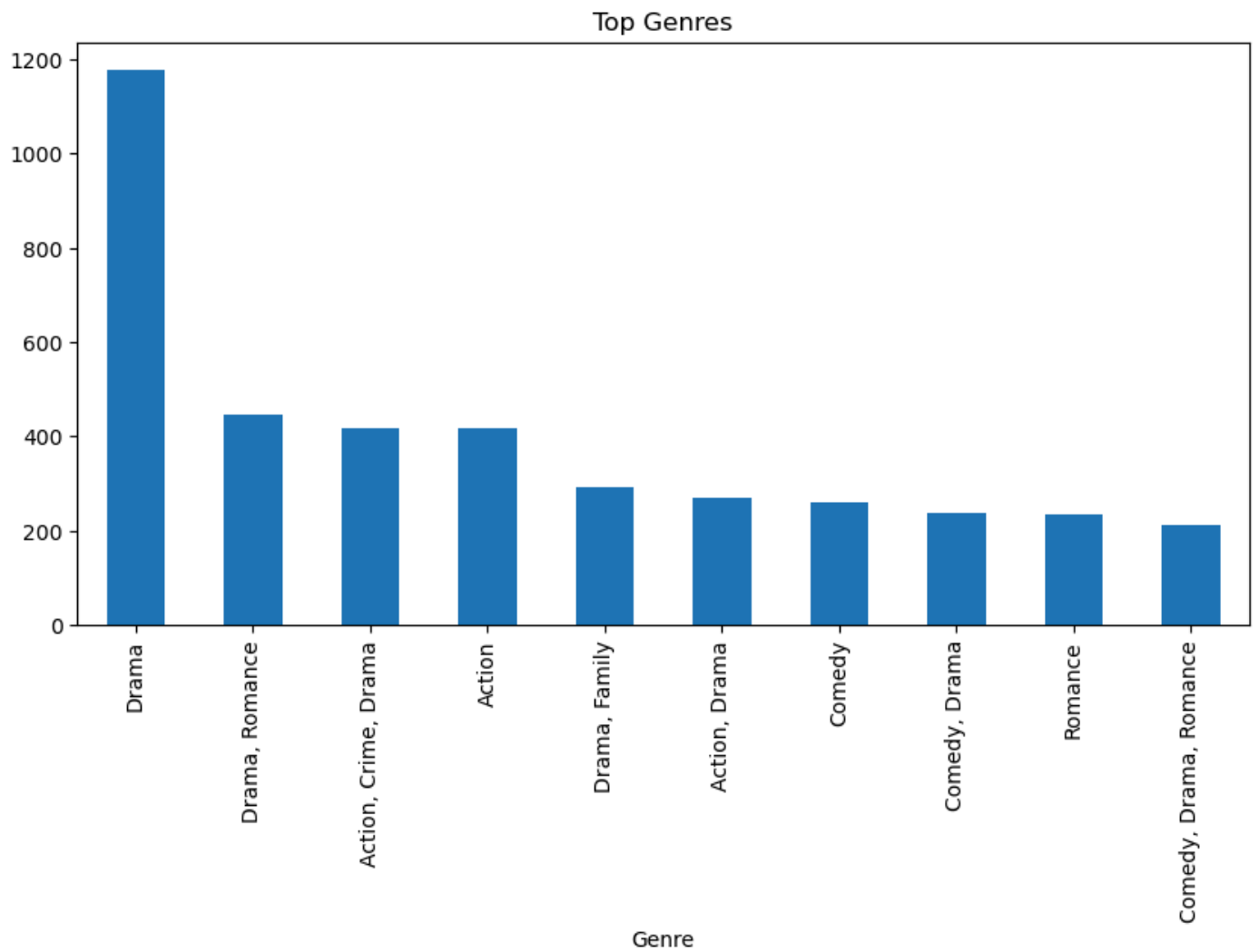


```
In [18]: plt.figure(figsize=(10,5))

df['Genre'].value_counts().head(10).plot(kind='bar')

plt.title("Top Genres")
plt.show()
```

Out:

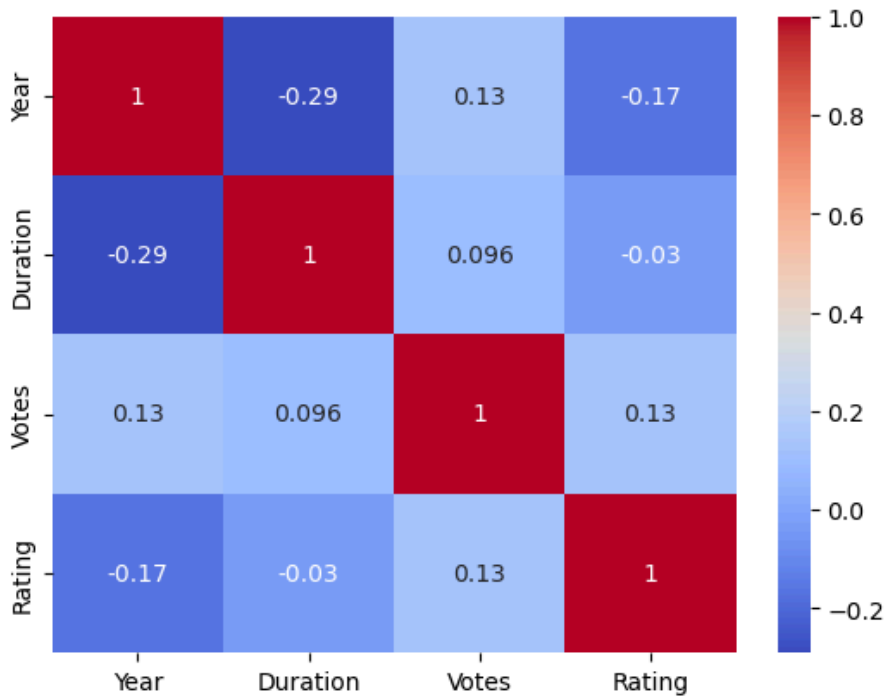


```
In [19]: numeric_df = df[['Year', 'Duration', 'Votes', 'Rating']]
```

```
sns.heatmap(  
    numeric_df.corr(),  
    annot=True,  
    cmap='coolwarm'  
)
```

```
plt.show()
```

Out:



```
In [20]: from sklearn.preprocessing import LabelEncoder
```

```
le_genre = LabelEncoder()
le_director = LabelEncoder()
```

```
le_actor1 = LabelEncoder()
le_actor2 = LabelEncoder()
le_actor3 = LabelEncoder()
```

```
df['Genre'] = le_genre.fit_transform(df['Genre'])
```

```
df['Director'] = le_director.fit_transform(df['Director'])
```

```
df['Actor 1'] = le_actor1.fit_transform(df['Actor 1'])
```

```
df['Actor 2'] = le_actor2.fit_transform(df['Actor 2'])
```

```
df['Actor 3'] = le_actor3.fit_transform(df['Actor 3'])
```

```
In [21]: print(df.columns.tolist())
```

Out:

```
['Name', 'Year', 'Duration', 'Genre', 'Rating', 'Votes', 'Director', 'Actor 1', 'Actor 2', 'Actor 3']
```